Dylan Craver

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Education

University of California, San Diego Expected Graduation: 2027
Bachelor of Science in Data Science

Executive Summary

I am a collaborative and highly communicative problem solver with a creative approach to data analysis, visualization, and feature engineering. I enjoy working closely with others to transform raw data into insights that improve processes and support impactful decisions.

Technical Skills

Data Analysis : Tableau and Power Bi (basic), Analytical Skills (Root Cause Analysis), EDA, Statistical Testing
Machine Learning: TensorFlow, PyTorch, scikit-learn, XGBoost, Feature Engineering, Model Evaluation, Cross-Validation, Hyperparameter Tuning, Fairness Analysis, Deep Neural Networks (CNNs, RNNs, LSTMs)
Data Collection: Web Scraping (Selenium, BeautifulSoup), API Integration
Programming: SQL, Python (Pandas, NumPy, Plotly, Matplotlib)
Tools: Git, GitHub, Jupyter Notebook, Excel, Databricks, Azure,

Relevant Coursework

Computer Science: Foundations of Computer Science, Problem Solving with Computers, Data Structures and Algorithms
Data Science: Modeling and Data Analysis, Practice and Application of Data Science, Theoretical Foundations of Data Science
Math/Statistics: Multi-variable Calculus, Linear Algebra with Applications, Statistical Methods,

Projects

Machine Learning Analysis of U.S. Power Grid Outages Nov 2025

- Analyzed 1,515 major outages and developed performance metrics to quantify outage severity, duration, and regional risk patterns.
- Identified data trends linking climate anomalies and population density to outage likelihood, generating insights to explain metric movement.
- Trained Machine Learning Algorithms (Logistic Regression and Decision Tree models) using scikit-learn Pipelines and GridSearchCV to evaluate predictive signals.
- Performed a 5,000-trial fairness analysis revealing statistically significant disparities across NERC regions ($p=0.0016$).

AI Powered Community College Transfer Planner Jul 2025 – Present

- Built an AI-driven academic pathway generator using dependency graphs to optimize student course sequencing and identify key bottlenecks.
- Designed ETL pipelines transforming articulation rules into structured data enabling fast metric lookup (GE progress, unit tracking, requirement completion).
- Created APIs to surface generated insights—such as completion timelines, requirement fulfillment metrics, and optimal-term recommendations.
- Developed visual interfaces enabling students and advisors to interpret academic metrics and make data-informed planning decisions.

College Research Automation Platform Jan 2025 – Present

- Developed an analytics platform for GetItWrite College Advising supporting data-driven decision-making for dozens of users.
- Built a data retrieval engine integrating College Scorecard API to analyze trends in tuition, acceptance rate, financial aid, and student outcomes.
- Created interactive dashboards highlighting key metrics and patterns for user-driven filtering and comparison.
- Implemented Google OAuth and automated Sheets workflows to surface personalized insights and reduce manual data processing time.